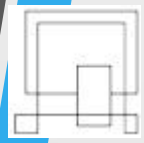
VIA University College



Programming 3

Autumn 2026

Learning Objectives

- ✓ explain the use of Maven, its advantages and disadvantages.
- ✓ explain maven POM file, GA(V) and other elements.
- ✓ create a new maven project from the command line
- ✓ compile and execute your Java code into an executable JAR file from the command line
- ✓ add Unit tests to the code and run the tests using maven commands.

What is Maven?

- A build tool for Java projects
- A management and automation tool
- It is written in Java
- Can be used to build and manage projects in:
  - Kotlin
  - C#
  - Scala
  - Ruby
  - Other languages

Maven in a Java Project

- POM file
  - A Project Object Model (POM stored in a **pom.xml** file) provides all the configuration for a single project.
- Plugins
  - Most of Maven's functionality is in plugins.
  - A plugin provides a set of goals that can be executed
- Build lifecycles
  - Build lifecycle is a list of named phases that can be used to give order to goal execution
- Dependencies
  - A central feature in Maven is dependency management
- Repository
  - A directory on your machine where the project jars, plugins are stored.
  - Types
    - Local Repository
    - Central Repository
    - Remote Repository
- Artifacts
  - An artifact is a file, usually a JAR, that gets deployed to a Maven repository.

Installation and use

- Maven can be used through
  - command prompt: **`mvn -v`**
  - IDE - IntelliJ

Maven pom.xml file elements

- project: root element of the pom.xml
- modelVersion: shows the model version
- groupId: describes the Id of the project (corresponds mostly to a reversed domain)
- artifactId: specifies the id of the project
  - Artifacts (jars, wars) are either produced or used by the project
- version: version of the artifact under given group
- Additional elements
  - Packaging
  - Build
  - Dependency.

Maven pom.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>dk.via.pro3</groupId>
    <artifactId>java-maven-project1</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>

    <build>
        <plugins>
            <plugin>
                <groupId>…</groupId>
                <artifactId>…</artifactId>
                <version>…</version>
                <configuration>
                    …
                </configuration>
            </plugin>
            …
        </plugins>
    </build>
    <dependencies>
        <dependency>
          …
        </dependency>
    </dependencies>

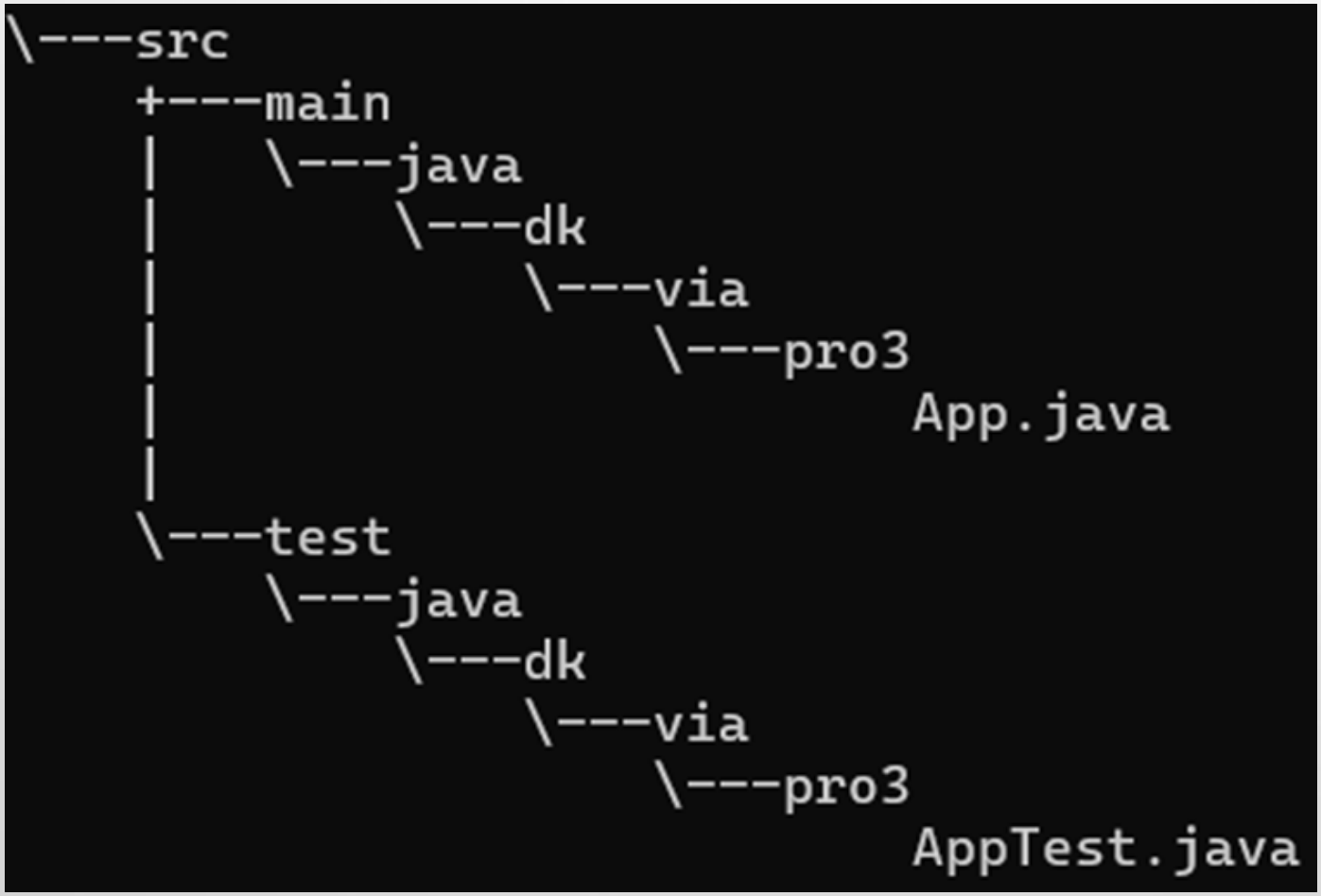

</project>
```

Advantages/Disadvantages

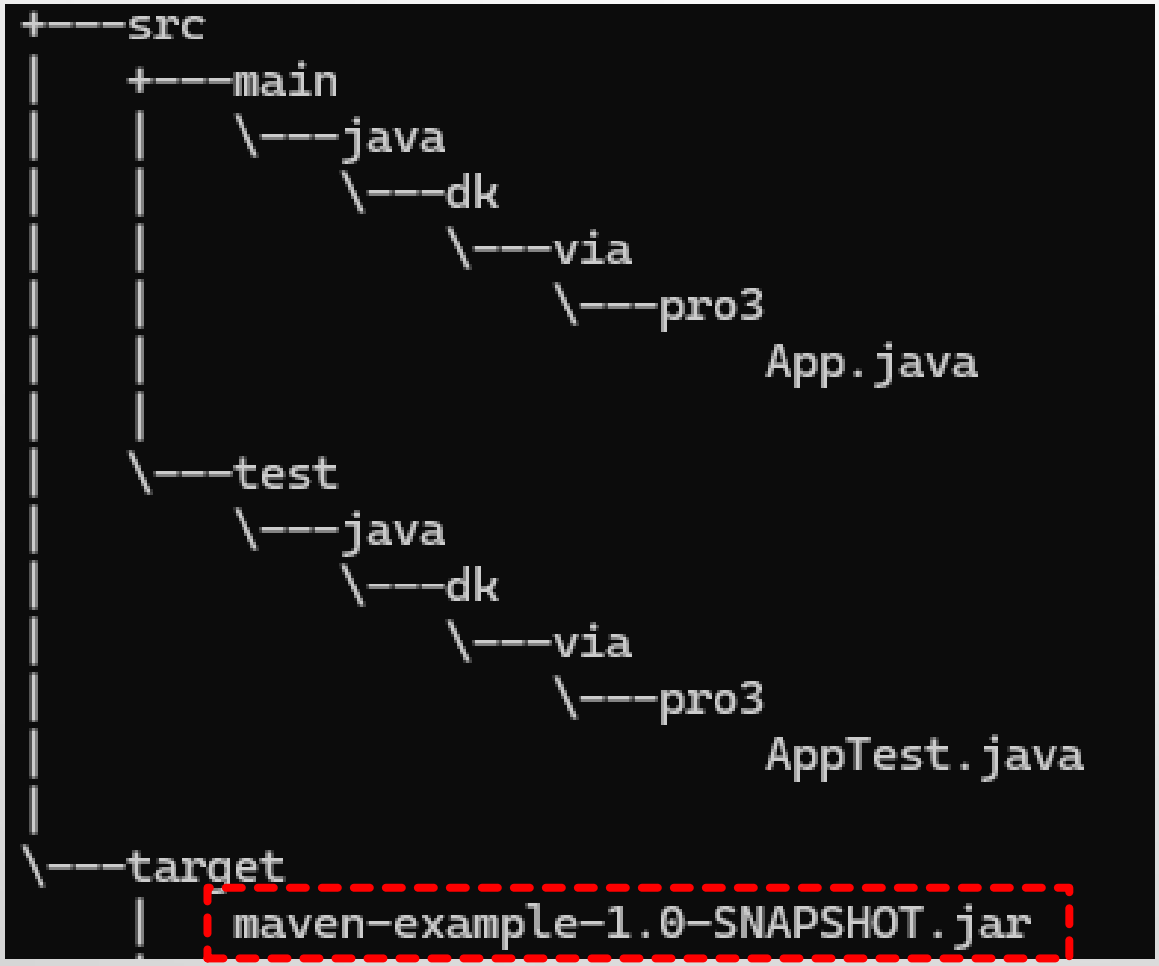
- ✓ Provides an easy start to the project in different environments
- ✓ Easy building of project to jar/war based on your requirements
- ✓ Easy to add new dependencies using the pom file.

- Dependency cannot be added if the maven code for an existing dependency is not available
- Must install Maven along with plugins before it can be used.

Maven Project Structure



After compiling the code



Live Coding/Hands-on/Demo

